

A Bayesian approach to prospective binary outcome studies with misclassification in a binary risk factor

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SUMMARY

Misclassification in a binary exposure variable within an unmatched prospective study may lead to a biased estimate of the disease–exposure relationship. It usually gives falsely small credible intervals because uncertainty in the recorded exposure is not taken into account. When there are several other perfectly measured covariates, interrelationships may introduce further potential for bias. Bayesian methods are proposed for analysing binary outcome studies in which an exposure variable is sometimes misclassified, but its correct values have been validated for a random subsample of the subjects. This Bayesian approach can model relationships between explanatory variables and between exploratory variables and the probabilities of misclassification. Three logistic regressions are used to relate disease to true exposure, misclassified exposure to true exposure and true exposure to other covariates. Credible intervals may be used to make decisions about whether certain parameters are unnecessary and hence whether the model can be reduced in complexity.

In the disease–exposure model, for parameters representing coefficients related to perfectly measured covariates, the precision of posterior estimates is only slightly lower than would be found from data with no misclassification. For the risk factor which has misclassification, the estimates of model coefficients obtained are much less biased than those with misclassification ignored. Copyright © 2005 John Wiley & Sons, Ltd.

KEY WORDS: errors in variables; binary outcome; measurement error; misclassification; odds ratio

1. INTRODUCTION

A prospective study may be used to examine the associations between a binary disease state and various exposures. A complication when investigating potential risk factors is misclassification of discrete exposure variables, which can occur due to forgetfulness or false

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reporting. Analyses that ignore this problem may give biased estimates and standard errors that are falsely small [1]. Often a higher proportion of subjects with disease (cases) are correct in their recollection of positive exposures than subjects without disease (controls) [2]. Differential misclassification can bias the effect of potential risk factors in either direction [3].

We consider a prospective study of unmatched subjects with a binary outcome that is to be analysed using a logistic regression model. We permit some covariate data to be measured at the same time as the data on disease state. We suppose that the model includes both categorical and continuous covariates and that there is misclassification in a binary variable. This variable is referred to as the exposure, but the methods could be applied to any binary variable.

Whilst retrospective studies are especially prone to misclassification errors, they also occur in prospective studies [4]. Among prospective studies, this is particularly true of cohort studies where not all of the exposure information is taken from records completed at the time of the exposures. Recently historic databases have become more commonly utilised to provide cohorts for investigating diseases that take a long time to develop (historical cohort studies [5]). These databases provide an opportunity to check exposure information which would otherwise be based on the subject's recall some time after the event. We should note that the retrospective gathering of information about exposure does not change a prospective study into a retrospective study. 'In a prospective study groups of patients with different characteristics are followed up to see whether an outcome of interest occurs' [5]. These previously disease-free subjects are followed forward in time. A retrospective case-control study is one in which 'a number of subjects with the disease in question (the cases) are identified along with some unaffected subjects (controls). The past history of these groups in relation to the exposure(s) of interest is then compared' [5]. In retrospective studies the proportion of subjects with the disease is often fixed and then covariate data are collected.

Information about exposure is usually more accurate when it is collected at the time that exposure occurred, but obtaining this information usually has a higher financial cost or requires more effort than standard sources of information, such as postal questionnaires. However the latter may be more reliant on memory and prone to error. The highest quality information is known as the 'gold standard' and we are interested in the situation where gold standard information is available for a random sample of subjects in an internal validation sub-study.

Matrix-based frequentist methods have been used effectively to correct for misclassification in a single categorical risk variable [6–8]. Other investigators have considered misclassification and mismeasurement in the presence of other perfectly measured covariates using frequentist methods. Assumptions such as a fixed error structure across the range of the explanatory variables [9] or non-differential measurement errors [10, 11] were made. For studies with internal validation, maximum likelihood methods had higher power and gave confidence intervals with more appropriate coverage probabilities than other frequentist methods [11].

Most Bayesian methods that employ a validation sub-study are quite general and allow differential measurement error in a single predictor [12–18]. If other perfectly measured covariates are considered, either there are only one or two of them, or else the covariates, both perfectly and imperfectly recorded, are restricted to be all categorical or all continuous [14].

Misclassification in a prospective binary disease outcome study including a validation sub-study with mixed continuous and categorical covariates has not been considered in detail by other authors. Seaman and Richardson [19] and Gustafson [18] describe the only Bayesian approaches to a similar problem.

We propose a Bayesian approach that uses a three part model: a disease model, an exposure model and a misclassification model, each of which takes a logistic structure. Uninformative priors are used throughout. Previous investigations have not permitted much flexibility in the relationships between covariates, although these will be appropriate with real data. We allow differential misclassification, dependence of misclassification parameters on the relevant perfectly measured exposure variable and other covariates, and relationships between the covariates.

Section 2 outlines the theory of our method. Section 3 presents the results of a real data example and a simulation based on real data using our methods. Section 4 discusses the results and outlines the implications for future work. The methods presented in this paper are simple to implement using MCMC methods in WinBUGS.

2. OVERVIEW OF MODELS

In this section we introduce some notation and describe the models that are considered.

2.1. Without misclassification

If there is no misclassification of categorical variables or measurement error in continuous variables, then a logistic model is commonly used to relate a binary disease state, D , to a number of exposures. For cases $D=1$ and for controls $D=0$. Let N denote the total number of subjects in the study. We suppose that there are S variables excluding disease state and the S -dimensional vector $\underline{X}$ is used to denote these variables. $\underline{X}_{\setminus j}$ is $\underline{X}$ with its j th component, X_j , deleted. Continuous variables are mean centred throughout as this helps avoid computational problems that can arise with highly correlated predictor variables.

The primary aim in most studies with a binary outcome is to estimate the odds ratio(s) relating the odds of the particular exposures among cases to the odds of these exposures among controls. We use the subscript i to denote the value of a quantity for subject i so D_i and $\underline{X}_i$ denote the disease state and the covariate values of subject i . When all S predictor variables are gold standard the probability of the i th subject being a case is $\Pr(D_i=1 | \underline{X}_i)$ which we denote as $\pi(D_i | \underline{X}_i)$. The function of the exposure variables is assumed to be linear unless there is evidence of an improved fit with an alternative function. $\mathcal{D}$ is defined as the full set of D_i 's for $i=1, \dots, N$. We assume

$$\text{logit}(\pi(D_i | \underline{X}_i)) = \lambda_0 + \underline{\lambda}'_1(\underline{X}_i) = \lambda_0 + \lambda'_{11}X_{1i} + \underline{\lambda}'_{\setminus 1}\underline{X}_{\setminus 1i} \quad (1)$$

for $i=1, \dots, N$ and some constants λ_0 and $\underline{\lambda}_1$. We use $\underline{\lambda}$ to denote the parameters λ_0 , $\underline{\lambda}_1$ collectively, λ_{11} to denote the first element of $\underline{\lambda}_1$ and $\underline{\lambda}_{\setminus 1}$ to denote the remaining $S-1$ elements. The baseline prevalence of disease is denoted $\pi(D_i | \underline{X}_0)$ where $\underline{X}_0$ takes reference values for all variables within $\underline{X}$.

For frequentist methods, prospective and retrospective analyses are largely equivalent [19, 20]. However, Bayesian methods for retrospective studies are more complex because the full sampling distribution is required [18, 19, 21].

2.2. With misclassification

2.2.1. Background. Following Richardson and Gilks [15] we have adopted an approach based on a disease model, a measurement model and an exposure model. This approach was used for studies with a validation sub-study, by Muller and Roeder [14] for continuous variables and by Gustafson [18] for binary variables. Unlike Muller and Roeder [14] and Seaman and Richardson [19], we have considered prospective data. Hence we can use information about the exposure distribution directly and this leads to a much simpler analysis.

2.2.2. Disease outcome model. A joint prior, $P(\underline{\lambda}, \underline{\omega})$, must be set on the baseline log odds of disease, λ_0 , the rest of the disease model parameters, $\underline{\lambda}_1$, and the marginal distribution of $\underline{X}$, whose parameters we denote by $\underline{\omega}$. The joint posterior distribution of $\underline{\lambda}$, $\underline{\omega}$ is

$$P(\underline{\lambda}, \underline{\omega} | \mathcal{D}, \underline{X}) \propto P(\underline{\lambda}, \underline{\omega}) P(\mathcal{D}, \underline{X} | \underline{\lambda}, \underline{\omega}) \quad (2)$$

The second term on the right hand side may be represented as $P(\underline{X} | \mathcal{D}, \underline{\lambda}, \underline{\omega}) P(\mathcal{D} | \underline{\lambda}, \underline{\omega})$ or $P(\mathcal{D} | \underline{X}, \underline{\lambda}, \underline{\omega}) P(\underline{X} | \underline{\lambda}, \underline{\omega})$. The form chosen will depend on whether the study is retrospective or prospective. In a retrospective study, $P(\mathcal{D} | \underline{\lambda}, \underline{\omega})$, which is needed to evaluate $P(\underline{X} | \mathcal{D}, \underline{\lambda}, \underline{\omega})$, must be obtained by integrating out $\underline{X}$. Numerical evaluations of this integral can be computationally intensive. Seaman and Richardson [19] discretize all their continuous covariates so that this integral can be expressed as a sum. This allows more exposure variables to be fitted in the same computing time. The loss of information, although potentially small, is a limitation.

Our methods are designed for prospective studies and the prospective likelihood may be obtained without requiring such complex methods as those needed for retrospective study data. We are able to retain continuous covariates, unlike Seaman and Richardson [19], and for most real data problems the computing time will remain tolerable.

An underlying assumption is that the probability of being selected for the validation study is independent of the true exposure state and so true exposure may be considered ‘missing at random’ for those not selected. We have allowed the selection to depend on disease status only. Others have considered selection based on disease and surrogate exposure status [11, 22, 23]. We suppose that a random sample of n_1 subjects form a validation sub-study group for which gold standard and surrogate data are available. For the remaining subjects, who form the main study group, only surrogate data are available.

2.2.3. Exposure model. Suppose X_j is a possibly misclassified binary exposure. The perfectly classified value of the exposure is denoted by E_j which may depend on any of the elements of $\underline{X}_{\setminus j}$. $E_j = 1$ if the relevant subject is exposed and 0 otherwise. To define an exposure model, we assume

$$\theta(E_j | \underline{X}_{\setminus j}) = \Pr[E_j = 1 | \underline{X}_{\setminus j}] = \frac{\exp(\beta_0 + \underline{\beta}'_1 \underline{X}_{\setminus j})}{1 + \exp(\beta_0 + \underline{\beta}'_1 \underline{X}_{\setminus j})} \quad (3)$$

Collectively, the parameters β_0 and $\underline{\beta}_1$ are denoted as $\underline{\beta}$. The observed value of X_j , which we call the apparent exposure, is 1 for those apparently exposed and 0 for those apparently not exposed.

2.2.4. Misclassification model. It is assumed that the appropriate structure for the relationship between the covariates and misclassification probabilities is a linear combination of the covariates. Suppose k is the value of E_j for a particular subject ($k = 0$ or 1). For cases we assume the misclassification model has the form

$$\phi_k(X_j | \underline{X}_{\setminus j}) = \Pr[X_j = 1 | E_j = k, \underline{X}_{\setminus j}] = \frac{\exp(\gamma_k + \xi_k + (\underline{\eta}_k + \underline{\zeta}_k)' \underline{X}_{\setminus j})}{1 + \exp(\gamma_k + \xi_k + (\underline{\eta}_k + \underline{\zeta}_k)' \underline{X}_{\setminus j})} \quad (4)$$

Similarly, for controls we assume the misclassification model has the form

$$\psi_k(X_j | \underline{X}_{\setminus j}) = \Pr[X_j = 1 | E_j = k, \underline{X}_{\setminus j}] = \frac{\exp(\gamma_k - \xi_k + (\underline{\eta}_k - \underline{\zeta}_k)' \underline{X}_{\setminus j})}{1 + \exp(\gamma_k - \xi_k + (\underline{\eta}_k - \underline{\zeta}_k)' \underline{X}_{\setminus j})} \quad (5)$$

Then γ_k is the mean constant term for any subject with $E_j = k$. The $S - 1$ elements of $\underline{\eta}_k$ represent the relationship between misclassification and other covariates for subjects with covariate exposure $\underline{X}_{\setminus j}$. If there is differential misclassification, parameters ξ_k and $\underline{\zeta}_k$ are non-zero. If there is a relationship between misclassification and other covariates and this differs between cases and controls, then at least one of the $S - 1$ elements of $\underline{\zeta}_k$ will be non-zero. Since the constant term is $\gamma_k + \xi_k$ for cases and $\gamma_k - \xi_k$ for controls, these parameters are separable. Similarly $\underline{\eta}$ and $\underline{\zeta}$ are separable. For notational convenience we use $\underline{v} = (\gamma_1, \xi_1, \underline{\eta}_1, \underline{\zeta}_1, \gamma_0, \xi_0, \underline{\eta}_0, \underline{\zeta}_0)$ to denote the group of parameters that collectively relate to misclassification.

For subjects within the validation study, the probability of being exposed, $\Pr(E_j = 1 | \underline{X}_{\setminus j})$, is estimated from the probability of being apparently exposed and the appropriate probability of being apparently exposed, given the true exposure state,

$$\begin{aligned} \Pr(X_j = 1 | \underline{X}_{\setminus j}) &= \Pr(X_j = 1 | E_j = 1, \underline{X}_{\setminus j}) \Pr(E_j = 1 | \underline{X}_{\setminus j}) \\ &\quad + \Pr(X_j = 1 | E_j = 0, \underline{X}_{\setminus j}) \Pr(E_j = 0 | \underline{X}_{\setminus j}) \end{aligned} \quad (6)$$

2.2.5. Estimation. The three parts of the model are: the disease model in equation (1); the exposure model in equation (3); and the misclassification model in (4) and (5) for cases and controls, respectively. These are linked together by equation (6) which may differ for cases and controls. For notational convenience the S components of $\underline{X}$ may be rearranged so that the potentially misclassified variable is the first vector component X_1 and $\underline{X}_{\setminus 1}$ is perfectly measured. Vague priors are given to all of the parameters from the disease model, $\underline{\lambda}$, the exposure model, β_0 and $\underline{\beta}_1$, and the misclassification model, γ and ξ ; we suggest using independent normal priors with very small precision, such as 0.001 or 0.0001. If the continuous covariates are assumed to have additive effects in the disease model, their coefficients are uncorrelated with this prior distribution, but may be correlated in the posterior distribution.

Once the misclassification model has been included with the disease model and the exposure model, the joint distribution may be written as the product of all the model conditionals. For subjects in the main study, E_j is unknown but within each iteration it may be estimated using equation (6) from the current estimates of θ and ϕ for cases, or θ and ψ for controls. The predicted values of E_j in the main study are nuisance parameters of little importance in themselves, but they allow the main study data on exposures $\underline{X}$ to contribute to the estimation of the disease model parameters $\underline{\lambda}$.

Let the joint prior of the model parameters be defined as $f(\underline{\lambda}, \underline{\beta}, \underline{v})$. If the perfectly measured covariates, $\underline{X}_{\setminus j}$, are not treated as random variables then the likelihood of the data given the parameters $\underline{\lambda}, \underline{\beta}, \underline{v}$ may be written as

$$\begin{aligned} \Pr(\mathcal{D}, E_j, X_j | \underline{X}_{\setminus j}, \underline{\lambda}, \underline{\beta}, \underline{v}) &= \Pr(X_j | \mathcal{D}, E_j, \underline{X}_{\setminus j}, \underline{\lambda}, \underline{\beta}, \underline{v}) \Pr(\mathcal{D} | E_j, \underline{X}_{\setminus j}, \underline{\lambda}, \underline{\beta}, \underline{v}) \\ &\times \Pr(E_j | \underline{X}_{\setminus j}, \underline{\lambda}, \underline{\beta}, \underline{v}) \end{aligned} \quad (7)$$

The first term on the right hand side of the likelihood may seem more suitable for a retrospective study, but the misclassification equations differ according to whether the subject is a case or a control (equations (4) and (5)) so this is an appropriate form.

The additional subscript i is used to represent the value for the i th subject for each of the quantities defined in equations. Then the joint posterior distribution of $\underline{\lambda}, \underline{\beta}, \underline{v}$ is

$$\begin{aligned} f(\underline{\lambda}, \underline{\beta}, \underline{v} | \mathcal{D}, E_j, X_j, \underline{X}_{\setminus j}) &\propto f(\underline{\lambda}, \underline{\beta}, \underline{v}) \prod_i \Pr(X_{ji} | D_i, E_{ji}, \underline{X}_{\setminus ji}, \underline{\lambda}, \underline{\beta}, \underline{v}) \\ &\times \prod_i \Pr(D_i | E_{ji}, \underline{X}_{\setminus ji}, \underline{\lambda}, \underline{\beta}, \underline{v}) \prod_i \Pr(E_{ji} | \underline{X}_{\setminus ji}, \underline{\lambda}, \underline{\beta}, \underline{v}) \end{aligned} \quad (8)$$

An alternative form including $\underline{\omega}$ can be used if $\underline{X}_{\setminus j}$ are random variables. The posterior distribution is analytically intractable, but Markov Chain Monte Carlo (MCMC) methods can be used to obtain estimates and make inferences.

2.3. Model fitting, model complexity and non-differential models

The part of the analysis most prone to unstable estimates with poor precision is the misclassification model. This has the lowest power because it effectively uses only the validation data. A table of apparent *versus* true binary exposure by disease state will reveal the rates of misclassification. If these are low then the misclassification model will have to be very simple or it will become unstable. We generally recommend a forward modelling strategy since overfitting is a concern. The complexity possible for a robust model also depends on the size of the validation study. In particular, only very large non-linear effects of continuous covariates on misclassification can be detected with validation studies of the size used in practice.

Model convergence was investigated using Brooks, Gelman, Rubin (BGR) plots on the burn-in iterations [24], for which two chains must be run from different initial values, and by checking that MC error was less than 5 per cent of posterior SD on the remaining iterations as recommended by Best *et al.* [25]. The deviance information criterion (DIC) [26], is a Bayesian measure of model fit penalized for the effective number of parameters. We could not use this measure directly because it involves the evaluation of the likelihood at the posterior mean estimates of all parameters. For binary nodes, such as E_j and X_j , the appropriate values for this substitution are unclear. The sum of the mean posterior deviance and the model degrees of freedom is used to get approximate DIC values. The model with the smallest DIC indicates the closest fit to the observed data. However, with a large number of observations, the penalty has only a small influence on the DIC.

If there are no relationships, or only limited relationships, between classification rates and other covariates then the complexity of the model may be substantially reduced. A parameter

representing one of the relationships might be removed if, say, its 95 per cent credible intervals contained the value zero. Model choice via the assessment of whether a parameter is significant using credible intervals under a diffuse prior has been recommended by some authors [27–29], but not others [30].

In particular, the misclassification model may be simplified if the non-differential misclassification assumption is justified. Using the 95 per cent credible intervals obtained from fitting these models, if ξ is not significantly different to a zero vector and $\underline{\zeta}$ is not significantly different to a zero matrix then misclassification may reasonably be assumed to be non-differential. The number of parameters can then be reduced by removing ξ and $\underline{\zeta}$ before refitting the model.

Potentially, all the variables which are in the disease model could be included in the misclassification and exposure models as well. However, scientific plausibility must also play an important role in the modelling strategy. If linear combinations of covariates are to be investigated then the model may be reparameterized orthogonally to reduce the possibility of high autocorrelation. The disease, exposure and misclassification models may all be expanded to include interaction terms and quadratic or higher order terms. However, the limitation upon the complexity of the models is the quantity of available data and hence the precision of the estimates. The misclassification model is based upon the smallest number of observations and so is the least robust to the introduction of additional terms.

Our example data set has low rates of misclassification. Therefore we have only been able to fit simple models for the probabilities of misclassification and exposure and we have not been able to investigate non-linear effects of continuous variables or models other than those with additive effects.

In summary, our modelling strategy focuses on inspection of the 95 per cent credible intervals to determine not only the relationships between the outcome and covariates, but also whether the exposure or classification probabilities depend on other covariates and whether differential misclassification can be assumed. Models containing parameters with credible intervals including zero are rejected in favour of simpler ones. Two checks of convergence are recommended. An approximation to the DIC [26] is of only limited use for model selection if deviance is much greater than the penalty function.

In practice, there would be a single validation substudy of fixed size, but in order to assess the robustness of the estimates obtained by the proposed methods, we have looked at three different random samples of 100 subjects as the validation substudy. To each of these we have added a further 100 subjects to see how increasing the size of the validation substudy affects the parameter estimates and stability of the model.

3. EXAMPLE, COMPARISONS AND SENSITIVITY

3.1. Example

The methods proposed are applied to real data from a study relating hypertensive diseases during pregnancy to cardiovascular morbidity later in life [31]. Women were classed as having eclampsia/pre-eclampsia (extreme forms of gestational hypertension), gestational hypertension, or normal blood pressure in their first pregnancy based on maternity records taken at the time

(1951–1970). 1197 women from each of the three groups were selected from a maternity register. For this article all forms of gestational hypertension (GH) are grouped together. In a postal questionnaire in 1997, the women were asked to recall if they had GH in their first pregnancy. Here, data from past records and from questionnaires are taken as the ‘gold standard’ information and potentially misclassified surrogate data, respectively. Their answers differed from the gold standard records for 10 per cent of those without GH and 34 per cent of those with GH. The binary outcome considered here is whether or not subjects had ever had a doctor’s diagnosis of (long term) high blood pressure (by 1997). For 1210 women there were complete data on this outcome, age, smoking status (never, ex-smoker and current smoker), body mass index (BMI), GH status (gold standard) and GH status (questionnaire).

100 of the 1210 subjects were randomly selected and formed the validation data with complete data on everything. For the remaining 1110 women the gold standard data were ignored. This was done in order to reproduce the situation of interest where gold standard data are only gathered for a subset (perhaps by hand searching paper records) and are unknown and uncollected for the remainder. A benefit of this example is that the results from these data can be compared with those from complete data on all 1210 subjects.

The data are not from a traditional prospective cohort design because the proportion of subjects who were truly exposed to the risk factor (GH) was fixed in the population from whom the sample were drawn rather than taking a random sample. Therefore, it is not possible to obtain a completely unbiased estimate of the proportion with long-term high blood pressure in the general population of parous women. However, this does not prevent questions about the relationships between the probability of exposure and age and misclassification and age being addressed. The data are of very similar form to that of a prospective cohort study and permit the evaluation of the proposed approach as a method of correcting for misclassification in prospective data. The data are not retrospective because the proportion of subjects with the disease was not fixed in advance. Long-term hypertension was found in 54 per cent of subjects with GH and 27 per cent of subjects without it [31].

The regression coefficients in the disease model correspond to: λ_0 the constant, λ_{11} GH status, λ_{12} mean centred age (age-60 years), λ_{13} mean centred BMI (BMI-25.6 kg/m²), λ_{14} dummy current smoking and λ_{15} dummy ex-smoking. We assumed additive effects for the continuous covariates in the disease model. We also allowed the probabilities of exposure to GH and classification to depend on age and/or BMI, so $\underline{\beta}_1 = (\beta_{11}, \beta_{12})'$. Classification probabilities might additionally depend on true exposure to GH, so $\underline{\eta}_k = (\eta_{k1}, \eta_{k2})'$ and $\underline{\zeta}_k = (\zeta_{k1}, \zeta_{k2})'$.

To give baseline results for later comparison, we first consider the special case where (a) true exposures are known for all 1210 women and (b) all recalled exposures are treated as if they are without misclassification. As these situations do not require misclassification to be taken into account, it is straightforward to analyse them using standard Bayesian methods in WinBUGS to estimate $\underline{\lambda}$. A burn-in of 10 000 iterations and a further run of 20 000 iteration took 728 s. Results are given in Table I. The log odds ratios (bold type) and 95 per cent credible intervals (regular type) indicate that a doctor’s diagnosis of long term high blood pressure in later life was significantly associated with the presence of true GH exposure, increasing age, increasing BMI, and reduced for current smokers relative to non-smokers. The log odds ratio for true GH exposure on all 1210 subjects was 1.11 (95 per cent credible interval: (0.81, 1.42)). If the recalled data were used as if they were true, the point estimate for GH status was greatly inflated to 1.52 and the 95 per cent credible interval indicates that the relationship with smoking was non-significant. An alternative to correcting for misclassification

Table I. Posterior mean log OR* and 95 per cent credible intervals without misclassification.

	True all known $N = 1210$			Recall all known $N = 1210^\dagger$		
	log OR	2.5 %	97.5 %	log OR	2.5 %	97.5 %
Constant λ_0	-0.98	-1.29	-0.69	-0.95	-1.18	-0.73
GH λ_{11}	1.11	0.81	1.42	1.52	1.27	1.78
Age λ_{12}	0.02	0.01	0.04	0.04	0.02	0.05
BMI λ_{13}	0.09	0.06	0.12	0.09	0.06	0.12
Smoker λ_{14}	-0.37	-0.72	-0.02	-0.28	-0.64	0.08
Ex-smoker λ_{15}	0.16	-0.11	0.44	0.18	-0.11	0.47

*Calculated using WinBUGS 1.4.

 † Recalled data treated as complete and without misclassification.

would be to use only gold standard data in a smaller study. With only 100 subjects such wide 95 per cent credible intervals were found that they all spanned zero, including the interval for the presence of true GH exposure (0.97, 95 per cent CI: -0.05, 2.02).

Initially, an extremely flexible model was considered in which the probability of exposure and misclassification depend on both age and BMI. However, the diagnostic plot of Brooks and Gelman [32] indicated that the Markov chain did not converge in 10 000 iterations. Also, the parameter estimates were unstable and autocorrelation was close to 1 for some parameters, indicating that a model of this complexity is unsuitable for the given data. Instead, the following nine models were considered. They allow varying degrees of flexibility in the relationships between covariates.

Model 1: The probability of true exposure to GH is constant and the probabilities of being classified as exposed to GH are constant, but may differ between cases and controls. This requires that the parameters relevant to age and BMI in both the exposure model (β_1) and the misclassification model (η_k and ζ_k) are all set to zero.

Model 2a: The probability of true exposure to GH is allowed to depend on age, so $\beta_{11} \neq 0$. The probabilities of classification as exposed to GH are constant so all elements of η_k and ζ_k are set to 0.

Model 3a: The probabilities of classification as exposed to GH are allowed to depend on age, but the slope with age differs only between truly exposed and unexposed subjects regardless of disease state. $\eta_{k1} \neq 0$ and ζ_{k1} may take only two different values according to true exposure state.

Model 4a: As Model 3a, but the probabilities of classification as exposed to GH are allowed to depend on age and disease state so $\eta_{k1} \neq 0$ and $\zeta_{k1} \neq 0$.

Model 5a: Both the probability of true exposure to GH and probabilities of classification as exposed to GH are allowed to depend on age.

Models 2a, 3a and 4a are nested within model 5a. Model 3a is nested within model 4a and has 2 fewer parameters. Model 1 is the simplest model and is nested within all other models. We also define Models 2b–5b to be the same as Models 2a–5a, but with BMI as the continuous covariate instead of age.

Models were fitted to the data using the Bayesian methods described in Section 2. MCMC methods were applied using WinBUGS with two chains of 30 000 iterations (including 10 000

burn-in) starting from different initial values. Throughout, λ_0 and elements of $\underline{\lambda}_1$ were given normal priors with mean 0 and precision 0.0001 and other parameters were given normal priors with mean 0 and precision 0.001.

We wish to address certain questions with these data. (1) Can our method produce reasonable estimates and intervals for the disease model parameters? (2) Is there differential misclassification in these data? (3) Is there dependence on age or BMI in the exposure or classification probabilities?

For (1) we see if we are able to produce point estimates from the misclassification data that are close to those from the complete data analysis of 1210 subjects. For (2) and (3) we examine whether credible intervals for the appropriate parameters include the value 0.

3.2. Analysis based on real data

The relationship between true exposure to GH and classification according to recalled exposure is shown in Table II for the entire study and in the validation study data. Classification rates differ between the validation study and the whole study, as will generally be the case. In the validation study data, the probabilities of being classified as exposed were quite different in truly exposed cases, 73 per cent, and truly exposed controls, 53 per cent. This suggested that misclassification was differential.

Table III shows results of five models. All allow differential misclassification, as the parameter ξ_k can differ between cases and controls. Since the 95 per cent credible interval for parameter ξ_k excluded 0 in all models, it is clear that appropriate models should allow differential misclassification. Results for λ_1 in all models indicated that the outcome was significantly associated with exposure to GH (λ_{11}), increasing age (λ_{12}), increasing BMI (λ_{13}), and the odds were slightly decreased in smokers (λ_{14}). The point estimates of most of the disease model parameters were unbiased with posterior SDs only slightly bigger than with complete data for all subjects. The exceptions were SDs for the constant and for exposure to GH (SD for $\lambda_{11} = 0.44$) which was much bigger than with complete data for all subjects (0.16), but smaller than that obtained from the validation data alone (0.53).

Models were compared using the deviance information criteria proposed by Spiegelhalter *et al.* [26]. Since the number of subjects in the main study was large, the deviances in all models were high (2000–3000). The model degrees of freedom ranged from 11 to 16 and

Table II. Counts in the entire study and in the validation study with misclassified exposure to high blood pressure in pregnancy.

Apparent exposure (A)	Entire study group				Validation study group			
	Cases ($D = 1$)		Controls ($D = 0$)		Cases ($D = 1$)		Controls ($D = 0$)	
	$E = 1$	$E = 0$	$E = 1$	$E = 0$	$E = 1$	$E = 0$	$E = 1$	$E = 0$
1	412	16	210	12	29	1	17	1
0	108	58	209	185	11	8	15	18
Percentage $A = 1^*$	79 %	22 %	50 %	6 %	75 %	11 %	53 %	5 %

*Percentage of each disease and true exposure group classified as exposed by recall, $A = 1$.

were too small to form a very effective penalty function for over-fitting. Of all the models examined Model 1 had the highest deviance of all and seems too simplistic, because other models showed significant coefficients for terms it did not contain.

Models 5a and 5b, included terms with 95 per cent credible intervals spanning 0 and so seem unnecessarily complex. In particular the relevant β_1 term included 0, indicating no significant relationship between the probability of exposure to GH and age or BMI, respectively. Both Models 2a and 2b showed some evidence of relationships between the probability of exposure to GH and a continuous covariate (age or BMI, respectively), since the 95 per cent credible interval for relevant β_1 term excluded 0. However, Models 4a and 4b had smaller deviances than Models 2a and 2b, respectively. This implied that models with age or BMI in the misclassification model fitted these data better than models with age or BMI in the exposure model.

Model 4a had substantially higher deviance than Model 4b and so only Model 4b was investigated further. Inspection of its credible intervals revealed that the slope with BMI differed for exposed and non-exposed subjects. However, disease status had only a small influence on the slope so the simpler Model 3b was fitted instead. For patients with a BMI of 25.6 kg/m², Model 3b predicts the percentages of subjects reporting apparent exposure as close to 75 per cent for exposed cases, 50 per cent for exposed controls, 20 per cent for unexposed cases, and 0–5 per cent for unexposed cases. In exposed subjects the percentage reporting exposure increased by 1 per cent for each additional 1 kg/m² of BMI. In subjects without true exposure BMI had no significant effect. In terms of the parameters, $\beta_1 = 0$, while $\eta_{k2} \neq 0$ and differed for all disease and exposure combinations. Also $\zeta_{22} \neq 0$ and indicated increased reporting with increasing BMI.

Clearly certain models performed better in terms of goodness of fit while others had less biased estimates for the variable affected by misclassification. There is a danger of overfitting with more complex models, especially where some parameters (e.g. those representing slopes) must be estimated from small subsets that correspond to particular combinations of disease and exposure states.

3.3. Validation sample effects, sensitivity and large misclassification probabilities

Analyses were replicated with three different selections of 100 subjects as the validation substudy. The results generally showed consistency, although some of the point estimates, particularly for λ_{11} , were not robust to quirks of the selected validation sample. This was due to there being few misclassified subjects, so that some estimates were unstable. For Model 3b, estimates and 95 per cent credible intervals for λ_{11} (relating to GH exposure) were 0.70 (0.11, 1.30), 0.79 (–0.10, 1.80) and 1.44 (0.73, 2.15).

We examined the effect of increasing the size of the validation sample from 100 to 200 or 300 subjects. Point estimates for the misclassification parameters became more stable and their posterior SDs decreased as the validation sample size increased. While most of the disease model parameters were not greatly affected, the coefficient λ_{11} became more precise. There was also much more consistency between point estimates obtained across different random selections of subjects for the validation sample. For example, when 100 additional subjects were added to each of the three validation samples mentioned above, the point and interval estimates of λ_{11} became 0.96 (0.48, 1.46), 1.09 (0.46, 1.75) and 1.44 (0.78, 2.13), respectively. The 95 per cent credible intervals include the point and interval estimate for the coefficient

Table III. Posterior mean log OR and 95 per cent credible intervals allowing differential misclassification.

	Model 1 Simple			Model 2b θ BMI dep.			Model 3b ϕ, ψ BMI dep.			Model 4b ϕ, ψ BMI dep.			Model 5b θ, ϕ, ψ BMI dep.		
	log OR	2.5 %	97.5 %	log OR	2.5 %	97.5 %	log OR	2.5 %	97.5 %	log OR	2.5 %	97.5 %	log OR	2.5 %	97.5 %
β_0	1.09	0.70	1.52	1.03	0.67	1.42	1.03	0.72	1.38	0.94	0.66	1.27	0.95	0.64	1.30
β_{12}				0.10	0.04	0.16							-0.00	-0.07	0.08
γ_1^*	-2.40	-4.32	-0.96	-2.92	-4.98	-1.45	-3.19	-5.36	-1.61	-3.32	-5.82	-1.67	-3.31	-5.62	-1.77
γ_2	0.79	0.50	1.12	0.86	0.57	1.19	0.86	0.57	1.20	1.01	0.66	1.38	1.00	0.67	1.38
η_{12}^*							-0.03	-0.35	0.17	-0.09	-0.46	0.17	-0.08	-0.44	0.19
η_{22}							0.10	0.04	0.17	0.16	0.07	0.26	0.15	0.01	0.26
$\xi_1^{*,\dagger}$	0.77	-0.99	2.34	0.60	-1.33	2.47	0.65	-1.37	2.48	0.59	-1.45	2.81	0.61	-1.47	2.82
ξ_2	0.73	0.44	1.04	0.67	0.37	0.97	0.58	0.30	0.86	0.36	0.03	0.71	0.39	0.05	0.75
$\zeta_1^{*,\dagger}$							0.04	-0.30	0.35	0.06	-0.33	0.38	0.06	-0.33	0.38
$\zeta_{12}^{*,\dagger}$							-0.09	-0.20	0.01	-0.09	-0.19	0.02	-0.09	-0.19	0.02
ζ_{22}													-1.43	-2.14	-0.73
Constant λ_0	-0.85	-1.57	-0.22	-0.98	-1.68	-0.31	-1.19	-1.82	-0.61	-1.50	-2.17	-0.88	-1.43	-2.14	-0.73
GH λ_{11}	1.01	0.21	1.86	1.18	0.33	2.03	1.44	0.73	2.15	1.85	1.09	2.61	1.76	0.89	2.55
Age λ_{12}	0.03	0.01	0.04	0.03	0.01	0.05	0.03	0.01	0.05	0.03	0.01	0.05	0.03	0.01	0.05
BMI λ_{13}	0.10	0.07	0.13	0.09	0.06	0.12	0.11	0.08	0.14	0.12	0.08	0.15	0.12	0.08	0.15
Smoker λ_{14}	-0.42	-0.78	-0.06	-0.42	-0.79	-0.06	-0.40	-0.76	-0.03	-0.38	-0.76	0.00	-0.38	-0.76	-0.00
Ex-smoker λ_{15}	0.12	-0.17	0.40	0.12	-0.17	0.41	0.14	-0.15	0.43	0.16	-0.14	0.46	0.16	-0.15	0.46
Deviance (mean)	2884			2809			2731			2624			2646		

* A single subscript or the first of two subscripts refers to the true exposure state.
† Differential misclassification is assessed using the credible intervals of ξ and ζ .
 β_0 and β_1 are the constant and BMI dependent parameters within the exposure model.
 γ and η are the constant and BMI dependent parameters within the misclassification model.
 ξ relates to differences between cases and controls in the constant term within the misclassification model.
 ζ relates to differences between cases and controls in the BMI dependent term within the misclassification model.

for true GH exposure obtained from complete data analysis of all subjects. Similar findings were also observed for validation samples of 300.

To assess the sensitivity of Model 3 to the choice of prior distributions, other choices of prior were given to all stochastic parameters in the model. The first three choices were normal priors with a mean of zero and a precision equal to (i) 0.01, (ii) 0.001, (iii) 0.0001. The fourth choice was a t-distribution with a mean of zero, a precision of 0.001 and 10 degrees of freedom. The fifth choice was skew, having a normal prior with a precision of 0.01 below zero and 0.001 above zero. For all these priors, inspection of credible intervals indicated that differential misclassification and age dependence in the misclassification probabilities were appropriate. There was a little variation in the point estimate of λ_{11} , but 95 per cent credible intervals were of similar width regardless of the choice of prior. The overall interpretation of the estimates and intervals for disease model covariates were consistent regardless of the prior selected. Posterior estimates for misclassification parameters were less consistent. This confirms the findings of Richardson and Leblond [33] who found that (a) mis-specification of the prior distribution for the parameter relevant to the true exposure variable gave moderate bias in the estimate and larger posterior standard deviation for that parameter, and (b) prior mis-specification had very little influence on the parameter estimates of perfectly measured covariates.

Our study does not have a very large proportion of subjects with misclassification. Other articles have reported much larger proportions of misclassified exposures [2, 34, 35], where the consequences of ignoring the problem would have been greater. To demonstrate the effectiveness of our method in correcting for large rates of misclassification, we created data with 30 per cent of exposures misclassified in cases and 40 per cent in controls. When these data were used without correction, the relationship between the outcome and exposure to GH was halved in size (0.65, 95 per cent CI: 0.41, 0.89), but was still significant. Using Model 1 and a validation sample of 100, the posterior mean estimate of the log odds ratio was 1.06 (0.05, 2.09). Other, more complex models had higher deviance and credible intervals spanning 0 for parameters not in Model 1. These results suggest that the proposed method can recover the true parameter estimates for the disease model, even when misclassification rates are high.

4. CONCLUSIONS

A method has been developed here for analysing prospective data with misclassification in one covariate. We have been able to fit models including several covariates with complex relationships. The method is flexible and simple and is not sensitive to the choice of centrally peaked prior. The problem has not been tackled in this way before, partly because data of this type have not been readily available very frequently. However, in epidemiology the use of historical records combined with recent follow-up questionnaires is becoming much more popular. The method may also be easily adapted for use with a potentially misclassified categorical exposure variable. Using our method, we found that after correcting for misclassification, estimates of most regression coefficients and their posterior SDs were generally unbiased and almost as precise as those from an analysis of gold standard data on all subjects. The exception was for the misclassified binary exposure variable which had a posterior SD that was larger than with gold standard data for all subjects; the posterior SD was smaller than that obtained from analysing just the validation sample on its own, as might have been expected.

The methods given in this paper can all be implemented using WinBUGS 1.4 software and programs that implement the methods are given on our website (<http://mcs-notes2.open.ac.uk/Repository/StatsPublications1.nsf/>). For these data, chains of 30 000 iterations (including 10 000 burn-in) were run and the run-time for each chain was about 1474 seconds on a Pentium 4 PC. The processing time increased linearly with the number of subjects and was not prohibitively long. The examples in Section 3 illustrate that the proposed methods are viable ways of analysing prospective studies with binary disease outcomes.

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